QAOA Max-Cut example fails on GPU runners

https://github.com/NVIDIA/cuda-quantum/issues/623

ROW 14

QAOA Max-Cut example fails on GPU runners #623

Closed

bettinaheim opened on Sep 8, 2023 · edited by bettinaheim

Edits

Required prerequisites

- [x] Make sure you've read the documentation. Your issue may be addressed there.
- [x] Search the issue tracker to verify that this hasn't already been reported. +1 or comment there if it has.
- [ ] If possible, make a PR with a failing test to give us a starting point to work on!

Describe the bug

Enabling a basic validation that runs all examples on the GPU backends, the qaoa_maxcut.cpp example throws a runtime error due to lack of progress. Locally, everything works fine, but on the GH runners I am getting:

```
Validating qaoa_maxcut.cpp:
terminate called after throwing an instance of 'std::runtime_error'
  what():  NLOpt error: round-off errors limited progress (nlopt roundoff-limited).
Consider changing the stopping criteria (e.g., reducing max_eval or increasing f_tol) and/or increase the numb
Aborted (core dumped)
```

The qaoa_maxcut_builder.cpp example works fine, but that one is using a different optimizer.

Steps to reproduce the bug

In .github/workflows/publishing.yml enable the /home/cudaq/examples/cpp/other/gradients.cpp example in the validation job, then manually run the workflow for a recent deployment. It would be good to find a local repro, but I don't have one right now.

Expected behavior

The example should succeed.

Is this a regression? If it is, put the last known working version (or commit) here.

Not a regression

Environment

- **CUDA Quantum version**: current main
- **Python version**: N/A
- **C++ compiler**: any
- **Operating system**: N/A

Suggestions

No response

bettinaheim mentioned this on Sep 8, 2023

Add minimal validation of gpu backends and revert the NGC location #620

Gradients example doesn't make progress on the GPU runners #622

1tnguyen mentioned this on Sep 27, 2023

Support Volta+ arch in custatevec backend #696

Support Volta+ arch in custatevec backend #696

Merged 1tnguyen merged 2 commits into `NVIDIA:main` from `1tnguyen:tnguyen/fix-custatevec-bug` on Sep 27, 2023

Conversation 8 | Commits 2 | Checks 0 | Files changed 2

1tnguyen commented on Sep 27, 2023 Collaborator

Description

We fixed the `-gencode` option to `sm_80` (Ampere/A100) while the `CuStateVecCircuitSimulator.cu` does have some CUDA kernels (e.g., `__global__ void initializeDeviceStateVector`). Hence, the generated binary is not compatible with the V100 GPU.

In this case, the `initializeDeviceStateVector` was not executed since the runtime couldn't find a compatible binary to run; hence the state vector was left uninitialized (all 0.0's).

Fixed by modifying the `-gencode` to support more architectures: same level of compatibility as the cuquantum library.

Should fix the issues reported in #623; #622; #583

`support Volta+ arch in custatevec, same as cuquantum` 9b29668